



Measurements and thermodynamic modeling of the ethanol–water system with emphasis to the azeotropic region

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ABSTRACT

Recovery of bioethanol from the fermentation broth is classically achieved by distillation. The design and optimization of the distillation requires a reliable thermodynamic model that describes sufficiently the properties of the ethanol/water mixture, and, especially, the vapor–liquid equilibrium (VLE) near the azeotropic point. The purpose of this study is twofold: first, to provide new reliable low-pressure VLE data for the ethanol/water mixture with emphasis given near the azeotropic region, and second, to develop new temperature-dependent binary interaction parameters for three excess Gibbs energy models (UNIQUAC, Wilson and NRTL), by fitting simultaneously VLE data, azeotropic points and excess enthalpies for this mixture.

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1. Introduction

The continuously increasing energy demand leads to an increase in the consumption of fossil fuels. The exhaustion of such reserves, the increase of the cost of energy and the ecological repercussions, turn the interest to renewable sources of energy. One of these renewable energy sources is bio-ethanol, which is a liquid biofuel that can be produced from several biomass feedstocks and conversion technologies [1,2]. Bio-ethanol is an attractive alternative fuel because it is a renewable bio-based resource that provides the potential to reduce particulate emissions in compression–ignition engines [3,4].

The bio-ethanol production process consists of three main parts: (a) pre-treatment of biomass, (b) fermentation and (c) recovery of ethanol. During the last years many different processes have been examined for the recovery of near anhydrous ethanol. For cases in which fermentation products are more volatile than water, recovery by distillation is often the technology of choice. Distillation technologies that allow the economic recovery of dilute volatile products from streams containing a variety of impurities have been developed and commercially demonstrated [5].

There is a common problem in the dehydration of ethanol, because ethanol forms a minimum boiling azeotropic mixture with

water, which makes it impossible to completely separate ethanol from water in a single distillation column. Haelssig et al. [6] have simulated various schemes for the recovery of nearly anhydrous ethanol after the fermentation process. According to this work, distillation with two columns operating at different pressures represents the most economical and therefore, the most feasible option among the six alternatives considered (steam stripping and distillation, flash fermentation, single-column distillation, two-column distillation, distillation with heat pump and modified flash fermentation). The conclusion of their work agrees with the results of Larsson and Zacchi [7].

The accurate simulation of a distillation scheme that involves a series of columns working at atmospheric and sub-atmospheric pressures requires reliable experimental thermodynamic data as well as thermodynamic models. A lot of vapor–liquid equilibrium (VLE) data for the ethanol/water mixture as well as azeotropic points of this mixture have been presented in the literature. Also, various models have been used for the description of the phase equilibrium and other thermodynamic properties of the ethanol/water mixture. As far as the excess Gibbs energy models are concerned, Kolbe and Gmehling [8] fitted quadratic temperature-dependent interaction parameters for the Wilson and UNIQUAC models to a large database of ethanol/water phase equilibria data, while Boukouvalas et al. [9] correlated simultaneously VLE and azeotropic data with the Wilson equation, which in turn was used in the simulation of a two-column distillation system for the recovery of near-anhydrous ethanol. The model of Kolbe and Gmehling

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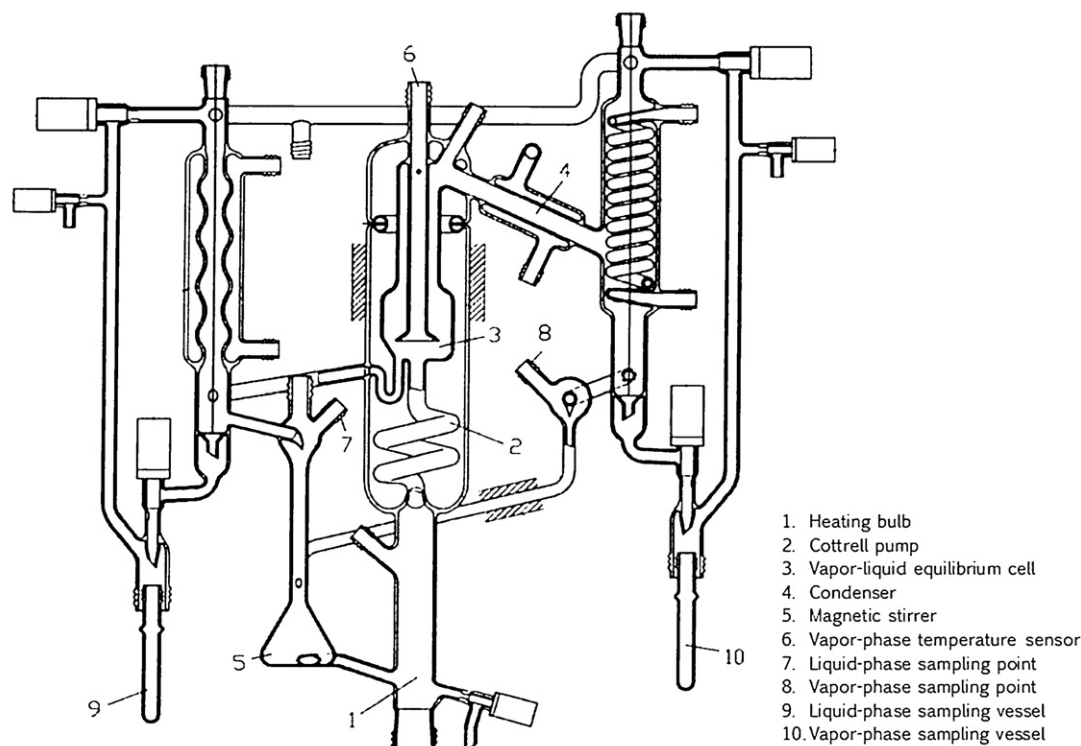


Fig. 1. Schematic diagram of the Labodest® VLE 602 (Fischer) ebulliometer used in this work.

fails to accurately predict the azeotropic points, especially for low pressures, while that of Boukouvalas et al., although it predicts fairly well the azeotropic points at low pressures, gives very poor heats of mixing predictions, which are useful information for the energy demand of the distillation process. It should be mentioned, of course, that besides the knowledge of the VLE behavior of the ethanol/water mixture, the accurate simulation and design of the recovery of ethanol from the fermentation broth, requires also the knowledge of the VLE behavior of other compounds present as impurities in the fermentation broth such as methanol and acetaldehyde.

The first aim of this study is to provide new experimental isobaric vapor–liquid equilibrium (VLE) data for the ethanol/water mixture at three sub-atmospheric pressures: 13.15, 19.71 and 32.86 kPa. The second aim is to develop new temperature-dependent binary interaction parameters for three commonly used in process simulators excess Gibbs energy (G^E) models (UNIQUAC, Wilson and NRTL), by simultaneously fitting VLE data, azeotropic points and excess enthalpies (H^E).

2. Experimental

2.1. Materials

Ethanol (Analytical reagent grade, 99.99%) was purchased from Fischer Scientific, while water (HPLC grade) and the solutions used for the Karl-Fischer titration were purchased from Merck. All reagents were used without further purification.

2.2. Apparatus and procedures

The equilibrium apparatus used in the VLE measurements is a glass Labodest® VLE 602 (Fischer) ebulliometer composed of a heating cell, an equilibrium cell and two water-cooled condensers/coolers. A schematic diagram of the apparatus is shown in

Fig. 1. The Cottrell pump ensures that the liquid and vapor phases are in intimate contact during boiling and also in contact with the temperature sensing element. The desired pressure is set by a pressure controller and measured with an accuracy of ± 0.01 kPa. The heating load of the immersion heater is set at the appropriate level in order to ensure a sufficient reflux ratio. Temperature is determined by a Pt100 resistance sensor with an accuracy of ± 0.01 K.

In each experiment, the pressure was fixed and the heating and stirring system of the liquid mixture was turned on. The apparatus was operated at constant pressure until equilibrium was reached (constant temperature). Once equilibrium was achieved, approximately after 30 min, samples of liquid and condensed vapor were withdrawn from the sample tubes and their composition was determined using a Titroline-KF Karl-Fischer Titrator. The maximum uncertainty in the composition measurements is about ± 0.001 in mole fraction.

2.3. Results

Isobaric VLE measurements for the ethanol/water binary mixture were performed at 13.15, 19.71 and 32.86 kPa, and they are presented in Table 1 and graphically in Fig. 2. The thermodynamic consistency of the data was tested with the method presented by Fredenslund et al. [10], using a third-order Legendre polynomial. Although this method is more robust for isothermal data, the experimental isobaric data are considered consistent if the average absolute deviation in the vapor phase composition is lower than 0.01. In Table 2 the mean absolute deviations in pressure and in vapor phase composition are given, and they indicate that all experimental points are thermodynamically consistent. Azeotropic points determined by interpolation of experimental data are presented in Table 3. Interpolation was carried out with the UNIQUAC activity coefficient model [11], using specially fitted parameters to each dataset, and the virial equation for the fugacity coefficient with virial coefficients taken from the Hayden–O’Connell method [12].

Table 1

Experimental vapor–liquid equilibrium data for the binary system ethanol (1) + water (2) at 13.15, 19.71 and 32.86 kPa.

$P = 13.15 \text{ kPa}$			$P = 19.71 \text{ kPa}$			$P = 32.86 \text{ kPa}$		
$T \text{ (K)}$	x_1	y_1	$T \text{ (K)}$	x_1	y_1	$T \text{ (K)}$	x_1	y_1
324.53	0	0	333.00	0	0	344.57	0	0
313.71	0.117	0.461	323.53	0.076	0.407	335.16	0.063	0.356
310.70	0.241	0.567	320.52	0.141	0.510	332.54	0.104	0.463
309.53	0.383	0.632	318.05	0.260	0.592	330.96	0.150	0.494
308.82	0.492	0.668	317.04	0.390	0.632	329.26	0.248	0.556
308.36	0.590	0.714	316.35	0.499	0.679	326.77	0.454	0.662
308.10	0.656	0.737	315.90	0.599	0.706	326.10	0.581	0.696
307.70	0.771	0.812	315.55	0.688	0.760	325.89	0.657	0.732
307.54	0.840	0.855	315.29	0.799	0.826	325.67	0.760	0.792
307.56	0.864	0.874	315.26	0.805	0.828	325.53	0.839	0.850
307.51	0.897	0.904	315.19	0.841	0.854	325.51	0.854	0.868
307.46	0.908	0.910	315.15	0.883	0.888	325.46	0.913	0.913
307.50	0.911	0.914	315.16	0.913	0.915	325.43	0.931	0.931
307.48	0.934	0.934	315.13	0.931	0.932	325.42	0.932	0.934
307.45	0.947	0.947	315.13	0.945	0.945	325.45	0.965	0.963
307.44	0.971	0.969	315.14	0.960	0.957	325.65	1	1
307.46	1	1	315.14	0.964	0.964			
			315.23	1	1			

3. Thermodynamic modeling

Three commonly used excess Gibbs energy (G^E) models, UNIQUAC [11], Wilson [13] and NRTL [14], were used to correlate thermodynamic data of the ethanol/water binary system. The model parameters were fitted to an extensive data base that includes: (a) VLE data presented in Table 4, (b) azeotropic points presented in Table 5, and (c) excess enthalpies presented in Table 6.

Vapor phase non-idealities were calculated with the virial equation using virial coefficients from the Hayden–O’Connell method [12], while vapor pressures were obtained from the following equation:

$$P^s = \exp \left(a + \frac{b}{T} + c \ln T + dT^e \right) \quad (1)$$

with P^s in bar and T in K. The corresponding parameter values for ethanol and water are given in Table 7.

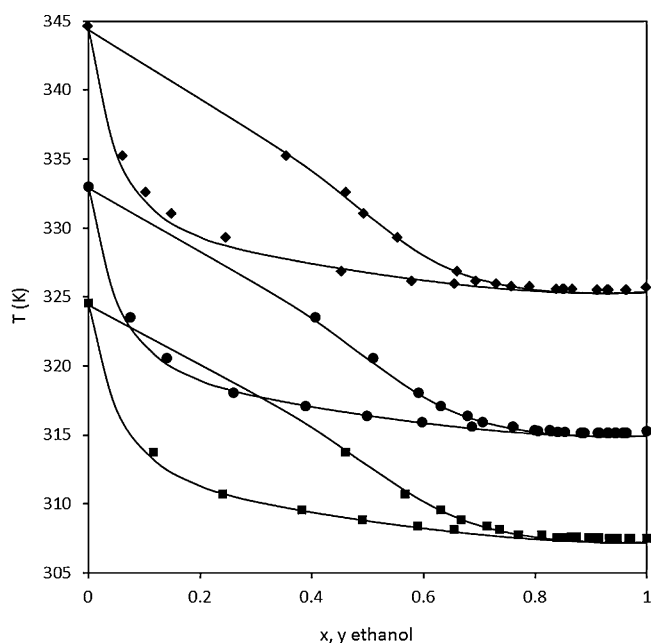


Fig. 2. Experimental vapor–liquid equilibrium data for the binary system ethanol (1) + water (2) at: (■) 13.15, (●) 19.71, and (◆) 32.86 kPa; solid curves correspond to the UNIQUAC model (this work) predictions.

Table 2

Results of the thermodynamic consistency test for the experimental VLE data measured in this study.

$P \text{ (kPa)}$	AADP ^a	AADy ^b
13.15	0.165	0.0055
19.71	0.197	0.0050
32.86	1.332	0.0095

$$^a \text{AADP} = \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left| p_i^{\text{exp}} - p_i^{\text{calc}} \right|$$

$$^b \text{AADy} = \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left| y_i^{\text{exp}} - y_i^{\text{calc}} \right|$$

NP is the number of the experimental data points.

4. Results and discussion

Quadratic temperature-dependent binary interaction parameters were determined for the three models by simultaneous fitting all the experimental data included in the database (Tables 4–6) using the maximum likelihood method. The objective function (OF) that was minimized for the regression of the database is the following:

$$\begin{aligned} \text{OF} = & \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left(\frac{T_i^{\text{exp}} - T_i^{\text{calc}}}{\sigma_T} \right)^2 + \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left(\frac{p_i^{\text{exp}} - p_i^{\text{calc}}}{\sigma_P} \right)^2 \\ & + \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left(\frac{y_i^{\text{exp}} - y_i^{\text{calc}}}{\sigma_y} \right)^2 + \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left(\frac{H_i^{\text{E,exp}} - H_i^{\text{E,calc}}}{\sigma_H} \right)^2 \\ & + \frac{1}{\text{NP}} \sum_{i=1}^{\text{NP}} \left(\frac{x_{\text{EtOH}}^{\text{azeotrope,exp}} - x_{\text{EtOH}}^{\text{azeotrope,calc}}}{\sigma_x} \right)^2 \end{aligned} \quad (2)$$

where σ s are the standard deviations in the determination of each property. The σ values were set equal to 0.1 for bubble point

Table 3

Azeotropic points determined in this study.

$P \text{ (kPa)}$	$T \text{ (K)}$	Ethanol mole fraction
13.15	308.05	0.954
19.71	315.61	0.938
32.86	325.56	0.918

Table 4

Experimental VLE database and correlation results for the ethanol/water mixture using UNIQUAC, Wilson and NRTL with the parameters determined in this work. Comparison with the corresponding results obtained with parameters proposed in the literature.

Data type	P (kPa)/ T (K)	UNIQUAC (this work)		NRTL (this work)		Wilson (this work)		UNIQUAC [8]		NRTL [15]		Wilson [9]		Ref. of exp. data
		$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	$\Delta T\%/\Delta P\%$	$\Delta y \times 100$	
Isobaric	50.66	0.09	0.72	0.04	0.43	0.06	0.54	0.05	0.39	0.07	0.42	0.16	1.21	[18]
Isobaric	101.32	0.18	1.64	0.07	0.56	0.05	0.80	0.05	0.54	0.07	0.43	0.18	1.22	[18]
Isothermal	298.14	0.97	0.59	0.65	0.46	1.24	0.80	1.03	0.69	3.17	2.09	2.02	1.51	[19]
Isothermal	298.15	1.17	0.91	1.03	0.72	1.51	1.12	1.17	0.80	4.12	2.07	2.16	1.54	[18]
Isothermal	313.15	1.63	0.55	1.88	0.50	2.45	0.79	1.76	0.47	3.53	1.56	2.34	0.78	[18]
Isothermal	323.15	2.04	0.89	1.88	0.98	2.53	1.26	2.30	0.95	3.06	1.45	3.74	1.71	[18]
Isothermal	328.15	0.44	0.72	0.93	0.60	1.36	0.59	0.60	0.61	0.81	0.87	1.43	0.53	[18]
Isothermal	343.15	0.38	0.74	0.82	0.68	1.25	0.52	0.97	0.55	0.87	0.56	1.92	0.60	[18]
Isothermal	348.15	1.40	0.15	0.73	0.83	0.48	0.92	0.42	0.67	0.48	0.62	1.34	1.31	[19]

$$\Delta T\% = \frac{1}{NP} \sum_{i=1}^{NP} \left| \frac{T_i^{\text{calc}} - T_i^{\text{exp}}}{T_i^{\text{exp}}} \right| \times 100; \Delta P\% = \frac{1}{NP} \sum_{i=1}^{NP} \left| \frac{P_i^{\text{calc}} - P_i^{\text{exp}}}{P_i^{\text{exp}}} \right| \times 100; \Delta y = \frac{1}{NP} \sum_{i=1}^{NP} |y_i^{\text{calc}} - y_i^{\text{exp}}|.$$

Table 5

Experimental azeotropic data and correlation results for the ethanol/water mixture using UNIQUAC, Wilson and NRTL with the parameters determined in this work. Comparison with the corresponding results obtained with parameters proposed in the literature.

P (kPa)	T_{azeo} (K)	x_{ethanol}	UNIQUAC (this work)		NRTL (this work)		Wilson (this work)		UNIQUAC [8]		NRTL [15]		Wilson [9]		Ref. of exp. data
			$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	$\Delta T_{\text{azeo}}\%$	$\Delta x_{\text{azeo}}\%$	
17.52	312.91	0.941	0.10	0.51	0.10	0.47	0.10	0.63	0.11	1.96	0.10	0.46	0.09	2.36	[20]
17.84	313.15	0.954	0.06	0.96	0.06	0.97	0.07	0.83	0.08	3.36	0.06	1.86	0.06	0.87	[9,21]
29.53	323.15	0.932	0.04	1.01	0.04	0.75	0.04	0.63	0.05	2.66	0.03	0.59	0.03	0.63	[22]
29.58	323.15	0.926	0.02	0.33	0.02	0.08	0.03	0.04	0.04	2.00	0.02	0.10	0.02	1.32	[9,23]
36.78	327.96	0.915	0.06	0.05	0.09	0.31	0.09	0.43	0.10	1.28	0.08	0.86	0.08	1.51	[20]
37.62	328.15	0.920	0.00	0.61	0.00	0.37	0.00	0.25	0.01	1.90	0.01	0.23	0.01	0.81	[9,21]
50.66	334.75	0.906	0.01	0.06	0.01	0.21	0.01	0.33	0.02	0.77	0.00	1.27	0.01	1.13	[9,24]
60.34	338.73	0.908	0.04	0.64	0.00	0.59	0.01	0.47	0.01	1.21	0.01	0.69	0.00	0.17	[25]
72.18	343.02	0.903	0.02	0.51	0.02	0.59	0.03	0.45	0.03	0.81	0.01	0.91	0.03	0.01	[25]
72.54	343.15	0.902	0.02	0.37	0.03	0.44	0.03	0.31	0.03	0.67	0.01	1.06	0.03	0.14	[9,26]
72.62	343.15	0.900	0.01	0.21	0.02	0.29	0.03	0.16	0.02	0.51	0.00	1.21	0.02	0.30	[22]
72.67	343.15	0.895	0.01	0.37	0.01	0.28	0.02	0.42	0.01	0.07	0.00	1.80	0.02	0.87	[9,21]
80.30	345.58	0.901	0.01	0.51	0.02	0.67	0.03	0.53	0.02	0.65	0.00	0.93	0.02	0.18	[25]
87.19	347.94	0.898	0.11	0.33	0.12	0.57	0.13	0.41	0.12	0.37	0.10	1.11	0.12	0.16	[20]
88.45	347.99	0.897	0.03	0.26	0.03	0.49	0.04	0.35	0.03	0.27	0.01	1.20	0.04	0.10	[25]
92.14	349.09	0.896	0.05	0.21	0.06	0.49	0.06	0.33	0.05	0.18	0.04	1.24	0.06	0.13	[25]
101.32	351.34	0.894	0.01	0.16	0.02	0.51	0.05	0.36	0.01	0.01	0.00	1.29	0.03	0.26	[25]
101.32	351.32	0.893	0.00	0.08	0.01	0.44	0.05	0.28	0.01	0.09	0.01	1.37	0.02	0.18	[22]
			0.03	0.40	0.04	0.47	0.05	0.40	0.04	1.04	0.03	1.01	0.04	0.62	

$$\Delta T_{\text{azeo}}\% = \left| \frac{T_{\text{azeo}}^{\text{exp}} - T_{\text{azeo}}^{\text{calc}}}{T_{\text{azeo}}^{\text{exp}}} \right| \times 100; \Delta x_{\text{azeo}}\% = \left| \frac{x_{\text{azeo}}^{\text{exp}} - x_{\text{azeo}}^{\text{calc}}}{x_{\text{azeo}}^{\text{exp}}} \right| \times 100.$$

Table 6

Excess enthalpy (H^E) correlation results for the ethanol/water mixture using UNIQUAC, Wilson and NRTL with the parameters determined in this work. Comparison with the corresponding results obtained with parameters proposed in the literature. The experimental data were taken from Larkin [27].

T (K)	x_{ethanol} range	ΔH^E (%)					
		UNIQUAC (this work)	NRTL (this work)	Wilson (this work)	UNIQUAC [8]	NRTL [15]	Wilson [9]
323.15	0.0365–0.9450	78	56	89	92	326	134
331.15	0.0302–0.9425	366	320	564	642	1456	607
343.15	0.0303–0.8957	75	83	115	179	522	247
363.15	0.0206–0.8833	35	65	55	316	982	636
383.15	0.0396–0.8833	11	20	31	45	147	131

$$\Delta H^E\% = \frac{1}{NP} \sum_{i=1}^{NP} \left| \frac{H_i^{E,\text{calc}} - H_i^{E,\text{exp}}}{H_i^{E,\text{exp}}} \right| \times 100.$$

Table 7

Values of the parameters of Eq. (1).

Compound	a	b	c	d	e
Ethanol	61.7911	-7.1223×10^3	-7.1424	2.8853×10^{-6}	2
Water	62.1361	-7.2582×10^3	-7.3037	4.1653×10^{-6}	2

temperatures, 1% for the bubble point pressures, 0.1% for the vapor and liquid phase concentrations and 10% for the excess enthalpy.

The parameters determined for the three models are listed in Table 8, while the correlation results in Tables 4–6. The results indicate that all three models, with the parameters determined in this work, provide similar and satisfactory results, with UNIQUAC being overall the best. Worse results are obtained by the three models in the description of the excess enthalpy. It should be noted that the excess enthalpy of the ethanol/water mixture shows a complex behavior, rendering it a good test system for checking whether G^E models are able to reproduce this behavior correctly. The excess enthalpy of the ethanol/water mixture is negative at low temperatures but increases steadily and becomes positive between 50 and 110 °C, depending on the concentration range (Fig. 3). The models considered in this work are able to reproduce qualitatively correctly this behavior, with the UNIQUAC model to give the best overall results (Fig. 3).

For comparison purposes we have examined the performance of the three models using parameters presented for these models in the literature: the UNIQUAC model with parameters from Kolbe and Gmehling [8], the Wilson model with parameters from Boukouvalas et al. [9], and the NRTL model with parameters used by NREL [15]. The results with these three models are also presented in Tables 4–6. The comparison of the models with the parameters developed in this work to those presented in the literature indicates a significant superiority of the former, especially in the azeotropic data (Fig. 4) and excess enthalpy description. The Wilson model using the parameters developed by Boukouvalas et al. [9] gives satisfactory azeotropic point and VLE predictions but very poor excess enthalpy predictions. The UNIQUAC model with parameters from Kolbe and Gmehling [9] fails in the azeotropic data predictions,

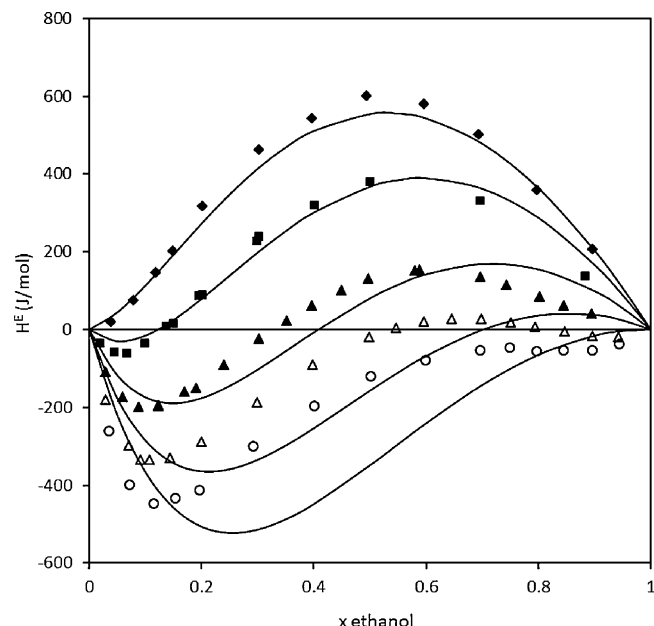


Fig. 3. Excess enthalpy for the ethanol/water mixture at various temperatures. (○) 323.15 K, (△) 331.15 K, (▲) 343.15 K, (■) 363.15 K, (◆) 383.15 K; solid curves correspond to UNIQUAC (this work) results.

while the NRTL model with the parameters used by NREL [15] gives the worst results.

The new parameters for the three models were tested against the VLE data measured in this work. Excellent predictions were obtained, which are shown in Table 9, with the UNIQUAC model to give the best predictions.

Finally, the performance of the UNIQUAC parameters determined in this work was also evaluated in VLE predictions of the ethanol/water mixture at high temperatures and pressures. To this purpose the Peng–Robinson Equation of State was coupled with the UNIQUAC model, using the parameters of Table 8, through

Table 8

New temperature-dependent interaction parameters for the UNIQUAC, NRTL and Wilson models.

Model	Interaction parameter expression	Parameter (K)	a	b	c
UNIQUAC ^a	$\frac{\Delta u_{ij}}{R} = a + bT + cT^2$	$\frac{\Delta u_{12}}{R}$	1030.38	−4.4853	0.004108
		$\frac{\Delta u_{21}}{R}$	−1881.18	9.2619	−0.009380
NRTL ^b	$\frac{\Delta g_{ij}}{R} = a + bT + cT^2$	$\frac{\Delta g_{12}}{R}$	−508.37	3.3910	−0.005840
		$\frac{\Delta g_{21}}{R}$	−763.53	5.1484	−0.003320
Wilson ^c	$\frac{\Delta \lambda_{ij}}{R} = a + bT + cT^2$	$\frac{\Delta \lambda_{12}}{R}$	−2007.08	10.8519	−0.013084
		$\frac{\Delta \lambda_{21}}{R}$	−472.30	4.4808	−0.005162

^a van der Waals volume (R) and van der Waals area (Q) parameters: $R_{\text{ethanol}} = 2.1055$, $R_{\text{water}} = 0.92$, $Q_{\text{ethanol}} = 1.972$, $Q_{\text{water}} = 1.2$.

^b The non-randomness parameter of the NRTL model was set equal to 0.3.

^c Liquid molar volumes: $V_{\text{ethanol}} = 58.69$ ml/mol, $V_{\text{water}} = 18.07$ ml/mol.

Table 9
Prediction of the VLE data measured in this work.

VLE data P (kPa)	UNIQUAC (this work)		NRTL (this work)		Wilson (this work)	
	$\Delta T\%$	$\Delta y \times 100$	$\Delta T\%$	$\Delta y \times 100$	$\Delta T\%$	$\Delta y \times 100$
13.15	0.04	0.35	0.20	1.70	0.12	0.93
19.71	0.04	0.34	0.22	1.70	0.11	0.74
32.86	0.06	0.70	0.24	2.07	0.10	1.03

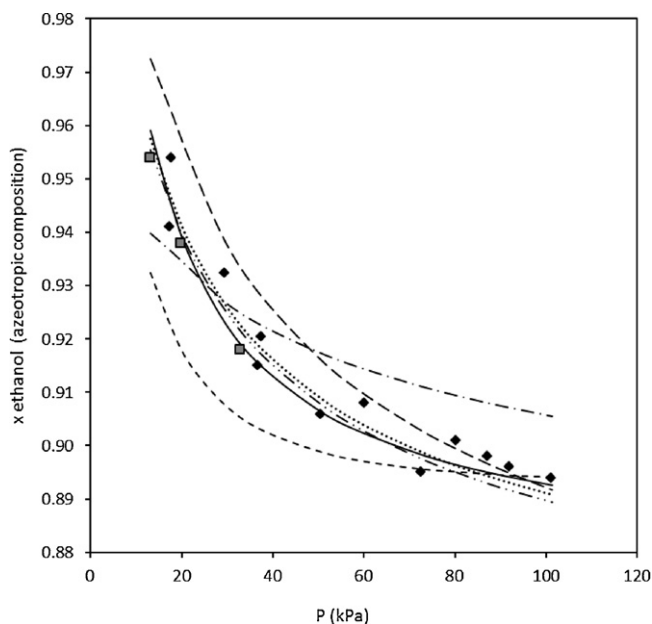


Fig. 4. Azeotropic ethanol compositions for the ethanol/water binary as a function of pressure. (♦) exp. data [9,20–26], (■) exp. data determined in this work, (—) UNIQUAC (this work), (····) Wilson (this work), (— · —) NRTL (this work), (---) UNIQUAC [8], (---) Wilson [9], (--- · ---) NRTL [15].

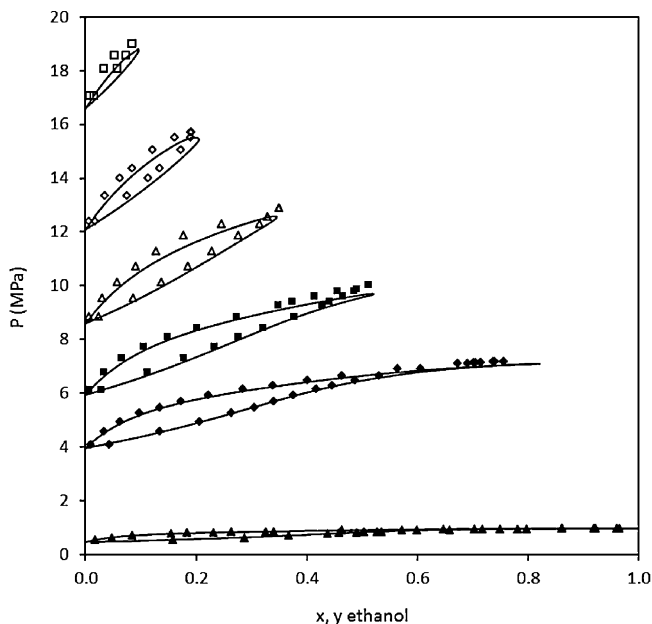


Fig. 5. VLE predictions for the binary system ethanol/water at high temperatures with the UMR/UNIQUAC model. (▲) 423 K, (◆) 523 K, (■) 548 K, (△) 573 K, (◇) 598 K, (□) 623 K, solid lines correspond to the model predictions. Experimental data were taken from Ref. [28].

the Universal Mixing Rules (UMR), which were proposed by Voutsas et al. [16,17]. The VLE predictions with the UMR/UNIQUAC model are presented in Fig. 5, and they indicate the very good performance of the new parameters even at temperatures and pressures well above the corresponding ranges of the database used for their fitting.

5. Conclusions

In this study the thermodynamic behavior of the binary ethanol+water was examined. New, thermodynamically consistent, isobaric VLE data were measured at 13.15, 19.71 and 32.86 kPa. The UNIQUAC, Wilson and NRTL excess Gibbs energy models were used for the simultaneous correlation of VLE data, azeotropic points and excess enthalpy data. The parameters determined in this work, compared to others presented in the literature, improve considerably the performance of the models, especially in the azeotropic data and excess enthalpy description. Finally, the UNIQUAC model with the parameters determined in this work, coupled with the Peng–Robinson Equation of State through the Universal Mixing Rules, provides very satisfactory VLE predictions of the ethanol/water mixture at high temperatures and pressures.

List of symbols

H^E	excess enthalpy
R	gas constant
P	pressure
P^s	vapor pressure of pure component
T	temperature
x_i, y_i	compositions of the liquid and vapor phases of component i
$\frac{\Delta g_{ij}}{R}$	interaction parameter of the NRTL model
$\frac{\Delta u_{ij}}{R}$	interaction parameter of the UNIQUAC model
$\frac{\Delta \lambda_{ij}}{R}$	interaction parameter of the Wilson model

Superscript

cal	calculated value
exp	experimental value

Acronyms

EtOH	ethanol
UMR	Universal Mixing Rules
VLE	vapor liquid equilibrium
NP	number of experimental data points

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